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Course code

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1 Particle Decay and Rockets

1.1 a

Problem: The relative velocity v between two particles can be determined from the four-scalar product of their four-momenta,

$$P_1 \cdot P_2 = m_1 m_2 c^2 \gamma(v)$$

(where $\gamma(v) \equiv 1/\sqrt{1 - v^2/c^2}$). Show this formula and motivate well.

Solution:

1.2 b

Problem: Consider a particle A at rest, decaying into two particles B and C. Use suitable invariants to show that the velocity of B is determined by

$$\gamma(v_B) = \frac{m_A^2 + m_B^2 - m_C^2}{2m_A m_B}$$

Solution:

1.3 c

Problem: Imagine that we can fuel a rocket with an ideal fuel, all of whose mass is converted to energy, leaving only massless exhaust that is released straight backwards. A rocket, that initially is at rest, is packed with ideal fuel. The fuel makes up a fraction α of the loaded rocket's total mass. Derive the maximum velocity of the empty rocket after burning all the fuel, as a function of α . (*Answer:* $v = c \frac{1-(1-\alpha)^2}{1+(1-\alpha)^2}$. *Hint:* Consider the process as a “decay”.)

Solution: